

# PADHAISHURU

GATE Preparation Platform

## GATE XE

Engineering Sciences

### TREND-BASED MOCK PAPER

— 01 —

|                 |             |
|-----------------|-------------|
| Duration        | 3 Hours     |
| Total Questions | 65          |
| Maximum Marks   | 100         |
| Sections        | 10 subjects |

#### Difficulty Distribution

Easy: 26 questions (40%)  
Moderate: 30 questions (46%)  
Difficult: 9 questions (14%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.  
These mock papers are constructed using analysis of historical GATE  
PYQ patterns.  
Historical patterns do not guarantee future examination questions.

# GENERAL INSTRUCTIONS

GATE XE — Trend-Based Mock Paper

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## 1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

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## 2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
  - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
  - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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## 3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
  - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
  - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
  - NAT questions carry no negative marking.
  - MSQ questions carry no negative marking.
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## 4. How to Answer

- For MCQ: Select exactly one option.
  - For MSQ: Select one or more options. Partial credit is NOT awarded.
  - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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## 5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

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## 6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

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## 7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
  - The Question Booklet contains multiple pages. Verify the page count.
  - Immediately report any discrepancy in the Question Booklet to the invigilator.
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**Q1.** 1M [-0.33]

[Basic Electrical]

V =:

- (A) IR
- (B) I/R
- (C) R/I
- (D)  $I^2R$

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**Q2.** 1M [no -ve]

[Environmental Science]

PM2.5 particle size:

Enter your answer here

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**Q3.** 1M [-0.33]

[Engineering Mathematics]

If A is 2x2 with trace 5 and det 6, eigenvalues:

- (A) 2 and 3
- (B) 1 and 4
- (C) 3 and 3
- (D) 2 and 4

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**Q4.** 1M [-0.33]

[Material Science]

Thermoplastics:

- (A) Thermoset
- (B) Remoldable
- (C) Crosslinked
- (D) None

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**Q5.** 1M [-0.33]

[Environmental Science]

PM2.5 particle size:

- (A)  $<2.5\mu\text{m}$
- (B)  $<5\mu\text{m}$
- (C)  $<10\mu\text{m}$
- (D)  $<1\mu\text{m}$

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**Q6.** 1M [-0.33]

[Basic Electrical]

V =:

- (A) IR
  - (B) I/R
  - (C) R/I
  - (D)  $I^2R$
-

Q7.

1M [-0.33]

*[General Aptitude]*If  $x^2=16$ ,  $x =$ :

- (A) 4
- (B) -4
- (C)  $\pm 4$
- (D) 0

Q8.

1M [no -ve]

*[Basic Electrical]*

V =:

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*Enter your answer here*

Q9.

1M [-0.33]

*[Basic Electrical]*

V =:

- (A) IR
- (B) I/R
- (C) R/I
- (D)  $I^2R$

Q10.

1M [-0.33]

*[Fluid Mechanics]*

Bernoulli assumes:

- (A) Viscous
- (B) Inviscid
- (C) Compressible
- (D) Turbulent

Q11.

1M [no -ve]

*[Fluid Mechanics]*

Bernoulli assumes:

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*Enter your answer here*

Q12.

1M [no -ve]

*[Material Science]*

Thermoplastics:

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*Enter your answer here*

Q13.

1M [-0.33]

*[Environmental Science]*

PM2.5 particle size:

- (A)  $<2.5;\mu\text{m}$
- (B)  $<5;\mu\text{m}$
- (C)  $<10;\mu\text{m}$
- (D)  $<1;\mu\text{m}$

**Q14.**

1M [-0.33]

*[Computers and Programming]*

Binary search complexity:

- (A)  $O(n)$
- (B)  $O(\log n)$
- (C)  $O(n^2)$
- (D)  $O(1)$

**Q15.**

1M [-0.33]

*[Environmental Science]*

PM2.5 particle size:

- (A)  $<2.5\mu\text{m}$
- (B)  $<5\mu\text{m}$
- (C)  $<10\mu\text{m}$
- (D)  $<1\mu\text{m}$

**Q16.**

1M [-0.33]

*[Environmental Science]*

PM2.5 particle size:

- (A)  $<2.5\mu\text{m}$
- (B)  $<5\mu\text{m}$
- (C)  $<10\mu\text{m}$
- (D)  $<1\mu\text{m}$

**Q17.**

1M [-0.33]

*[Thermodynamics]*

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

**Q18.**

1M [-0.33]

*[Computers and Programming]*

Binary search complexity:

- (A)  $O(n)$
- (B)  $O(\log n)$
- (C)  $O(n^2)$
- (D)  $O(1)$

**Q19.**

1M [no -ve]

*[Basic Electrical]* $V =$ :

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*Enter your answer here*

**Q20.**

1M [-0.33]

*[General Aptitude]*If  $x^2=16$ ,  $x =$ :

- (A) 4
- (B) -4
- (C)  $\pm 4$
- (D) 0

**Q21.**

1M [-0.33]

*[Thermodynamics]*

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

**Q22.**

1M [-0.33]

*[Environmental Science]*

PM2.5 particle size:

- (A)  $<2.5\mu\text{m}$
- (B)  $<5\mu\text{m}$
- (C)  $<10\mu\text{m}$
- (D)  $<1\mu\text{m}$

**Q23.**

1M [no -ve]

*[Basic Electrical]* $V =$ :

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*Enter your answer here***Q24.**

1M [-0.33]

*[Material Science]*

Thermoplastics:

- (A) Thermoset
- (B) Remoldable
- (C) Crosslinked
- (D) None

**Q25.**

1M [-0.33]

*[Fluid Mechanics]*

Bernoulli assumes:

- (A) Viscous
- (B) Inviscid
- (C) Compressible
- (D) Turbulent

**Q26.**

1M [no -ve]

*[Environmental Science]*

PM2.5 particle size:

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*Enter your answer here***Q27.**

1M [-0.33]

*[Computers and Programming]*

Binary search complexity:

- (A)  $O(n)$
- (B)  $O(\log n)$
- (C)  $O(n^2)$
- (D)  $O(1)$

**Q28.**

1M [-0.33]

*[Thermodynamics]*

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

**Q29.**

1M [-0.33]

*[Material Science]*

Thermoplastics:

- (A) Thermoset
- (B) Remoldable
- (C) Crosslinked
- (D) None

**Q30.**

1M [no -ve]

*[Fluid Mechanics]*

Bernoulli assumes:

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*Enter your answer here***Q31.**

2M [no -ve]

*[Material Science]*

Thermoplastics:

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*Enter your answer here***Q32.**

2M [-0.66]

*[Material Science]*

Thermoplastics:

- (A) Thermoset
- (B) Remoldable
- (C) Crosslinked
- (D) None

**Q33.**

2M [-0.66]

*[Environmental Science]*

PM2.5 particle size:

- (A)  $<2.5\mu\text{m}$
- (B)  $<5\mu\text{m}$
- (C)  $<10\mu\text{m}$
- (D)  $<1\mu\text{m}$

**Q34.**

2M [-0.66]

*[Environmental Science]*

PM2.5 particle size:

- (A)  $<2.5\mu\text{m}$
- (B)  $<5\mu\text{m}$
- (C)  $<10\mu\text{m}$
- (D)  $<1\mu\text{m}$

**Q35.**

2M [-0.66]

*[Solid Mechanics]*

Tensile stress formula:

- (A)  $F/A$
- (B)  $F \cdot A$
- (C)  $F/A^2$
- (D)  $A/F$

**Q36.**

2M [-0.66]

*[Thermodynamics]*

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

**Q37.**

2M [no -ve]

*[Computers and Programming]*

Binary search complexity:

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*Enter your answer here***Q38.**

2M [no -ve]

*[Environmental Science]*

PM2.5 particle size:

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*Enter your answer here***Q39.**

2M [-0.66]

*[General Aptitude]*If  $x^2=16$ ,  $x =$ :

- (A) 4
- (B) -4
- (C)  $\pm 4$
- (D) 0

**Q40.**

2M [-0.66]

*[Solid Mechanics]*

Tensile stress formula:

- (A)  $F/A$
- (B)  $F \cdot A$
- (C)  $F/A^2$
- (D)  $A/F$

**Q41.**

2M [-0.66]

*[Computers and Programming]*

Binary search complexity:

- (A)  $O(n)$
- (B)  $O(\log n)$
- (C)  $O(n^2)$
- (D)  $O(1)$

**Q42.**

2M [-0.66]

*[Basic Electrical]* $V =$ :

- (A)  $IR$
- (B)  $I/R$
- (C)  $R/I$
- (D)  $I^2R$

**Q43.**

2M [no -ve]

*[Material Science]*

Thermoplastics:

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*Enter your answer here***Q44.**

2M [no -ve]

*[General Aptitude]*If  $x^2=16$ ,  $x =$ :

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*Enter your answer here***Q45.**

2M [no -ve]

*[General Aptitude]*If  $x^2=16$ ,  $x =$ :

- (A) 4
- (B) -4
- (C)  $\pm 4$
- (D) 0

**Q46.**

2M [-0.66]

*[Material Science]*

Thermoplastics:

- (A) Thermoset
- (B) Remoldable
- (C) Crosslinked
- (D) None

**Q47.**

2M [-0.66]

*[Thermodynamics]*

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

**Q48.**

2M [-0.66]

*[Thermodynamics]*

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

**Q49.**

2M [-0.66]

*[General Aptitude]*If  $x^2=16$ ,  $x =$ :

- (A) 4
- (B) -4
- (C)  $\pm 4$
- (D) 0

**Q50.**

2M [no -ve]

*[Basic Electronics]*

Diode used as:

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*Enter your answer here***Q51.**

2M [-0.66]

*[Computers and Programming]*

Binary search complexity:

- (A)  $O(n)$
- (B)  $O(\log n)$
- (C)  $O(n^2)$
- (D)  $O(1)$

**Q52.**

2M [-0.66]

*[Material Science]*

Thermoplastics:

- (A) Thermoset
- (B) Remoldable
- (C) Crosslinked
- (D) None

**Q53.**

2M [-0.66]

*[Solid Mechanics]*

Tensile stress formula:

- (A)  $F/A$
- (B)  $F \cdot A$
- (C)  $F/A^2$
- (D)  $A/F$

**Q54.**

2M [no -ve]

*[Thermodynamics]*

First law:

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*Enter your answer here***Q55.**

2M [-0.66]

*[Material Science]*

Thermoplastics:

- (A) Thermoset
- (B) Remoldable
- (C) Crosslinked
- (D) None

**Q56.**

2M [-0.66]

*[Basic Electrical]* $V =$ :

- (A)  $IR$
- (B)  $I/R$
- (C)  $R/I$
- (D)  $I^2R$

**Q57.**

2M [no -ve]

*[General Aptitude]*If  $x^2=16$ ,  $x =$ :

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*Enter your answer here***Q58.**

2M [-0.66]

*[Material Science]*

Thermoplastics:

- (A) Thermoset
- (B) Remoldable
- (C) Crosslinked
- (D) None

**Q59.**

2M [no -ve]

*[Material Science]*

Thermoplastics:

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*Enter your answer here*

**Q60.**

2M [no -ve]

*[Fluid Mechanics]*

Bernoulli assumes:

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*Enter your answer here***Q61.**

2M [-0.66]

*[Material Science]*

Thermoplastics:

- (A) Thermoset
- (B) Remoldable
- (C) Crosslinked
- (D) None

**Q62.**

2M [-0.66]

*[Fluid Mechanics]*

Bernoulli assumes:

- (A) Viscous
- (B) Inviscid
- (C) Compressible
- (D) Turbulent

**Q63.**

2M [-0.66]

*[Thermodynamics]*

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

**Q64.**

2M [-0.66]

*[Fluid Mechanics]*

Bernoulli assumes:

- (A) Viscous
- (B) Inviscid
- (C) Compressible
- (D) Turbulent

**Q65.**

2M [-0.66]

*[General Aptitude]*If  $x^2=16$ ,  $x =$ :

- (A) 4
- (B) -4
- (C)  $\pm 4$
- (D) 0